

- Application agility and integration
- No vendor lock-in, no rigid architectures or proprietary formats and components

An IoT system has both human and digital users. Humans typically interact with the IoT system via devices like smartphones, personal computers, tablets or specialised gadgets. In all cases, the IoT system provides some type of application that connects the human user with the rest of the IoT system.

The IoT solution allows users to:

- Connect, provision, and manage millions of distributed IoT devices and gateways with added security
- Simplify data flow management with intelligence and analytics at the edge
- Provide a comprehensive, centralised advanced analytics and data management platform with the ability to build or refine machine-learning models and push these models to the edge
- Put security in the forefront (with pre-integrated security and manageability across devices), along with access, authentication and applications, as well as data that is in-motion and at-rest
- Use application development, deployment and integration services

Open source development platforms and frameworks

IoT platforms are the support software that connect everything in an IoT system. An IoT platform facilitates communication, data flow, device management and the functionality of applications. These platforms are classified into two categories — enterprise IoT platforms and consumer IoT platforms. Enterprise IoT platforms are used in oil and gas, manufacturing, or asset management. Consumer IoT platforms are for smart home applications or wearables.

The following are some of the top open source development platforms for IoT based applications.

Kaa: Kaa is a flexible, multi-purpose platform for developing end-to-end IoT solutions and connected applications. It provides better communication, control and interoperation capabilities in connected products and smart devices. It is primarily based on flexible microservices, and easily adapts to almost any need and application.

Sitewhere: This facilitates ingestion, storage, processing and integration of device data. It provides connectivity of devices using different protocols like MQTT, SMQP, STOMP, etc; devices can even be added through self-registration, REST services or in batches. The platform is based on the microservices architecture and is designed for reliability, high throughput, low latency and stability.

DeviceHive: This facilitates the connectivity of devices to the cloud within a short interval of time, i.e., just a few minutes to allow streaming of data and command traversal. The platform is distributed under the Apache 2.0 licence and is highly scalable via containerisation. It supports Docker and Kubernetes.

Zetta: This is an API based IoT platform built on Node.js. It is regarded as a complete toolkit to generate HTTP APIs for devices. Zetta combines REST APIs, WebSockets and reactive programming to build data-intensive and real-time applications.

Thingsboard.io: This is used for data collection, processing, visualisation and device management. It supports all standard IoT protocols like CoAP, MQTT and HTTP, as well as cloud and on-premise deployments.

Other open source IoT platforms are DSA, Thinger.io, Mainflux, Thingspeak, etc.

Open source middleware

Middleware acts as an agent between the service provider (IoT devices)

and service consumer (enterprise applications). A software layer sits between applications and objects. It is a mediator interface to provide the interaction between the Internet and 'things'. It hides heterogeneity among the devices, components and technology of an IoT system. Middleware provides solutions to frequently encountered problems, such as interoperability, security and dependability.

The following are some of the top open source middleware platforms for IoT based applications.

Kaa: Kaa is platform-centric middleware. It manages an unlimited number of connected devices with cross-device interoperability. It performs real-time device monitoring, remote device provisioning and configuration, as well as collection and analysis of sensor data. It is a microservices based portable, horizontally scalable, highly available IoT platform. It supports on-premise, public cloud and hybrid models of deployment.

IoTSyS: This is platform-centric and industry-specific middleware. It uses IPv6 for non-IP IoT devices and systems. It is used in smart city/ smart grid projects, making automation infrastructure smart. IoTSys provides interoperable Web technologies for sensors and actuator networks.

DeviceHive: This is cloud agnostic, microservices based platform-centric middleware used for device connectivity and management. It has the ability to connect to any device via MQTT, REST or WebSockets. It supports Big Data solutions like ElasticSearch, Apache Spark, Cassandra and Kafka for real-time and batch processing.

Zetta: This is application-centric middleware. It is a service-oriented open source IoT platform built on Node.js combining REST API, WebSockets and reactive programming. It can run on cloud